Marks scored:

KATHMANDU UNIVERSITY
End Semester Examination [C]
November, 2018

Level : B.E.
Year : II

Course : MCSC 202
Semester : II

Exam Roll No. :

Time: 30 mins.

F. M. : 10

Registration No.:

Date : NOV 15 2018

SECTION "A"
[10 Q. × 0.5 = 5 marks]

Fill in the blank space(s) by writing most appropriate word(s) or symbol(s).

1. The number 204.020050 has _____ significant digits.

2. If E and ∇ be respectively the shift and backward difference operators, then $\nabla \equiv$ _____.

3. Order of convergence of Regula-Falsi method is _____.

4. Minimizing the sum of the squares of the errors is called _____ method.

5. Simpson's rule requires the division of whole interval into a (an) _____ number of subintervals.

6. Given the matrix $A = \begin{bmatrix} 2 & 4 & -5 \\ 1 & 7 & 2 \\ 3 & -1 & 6 \end{bmatrix}$, the infinity norm $\|A\|_\infty =$_____.

7. If E be the shift operator, then $E^4 y_1 =$ _____.

8. Consider an initial value problem, $y' = x + y^2$ subject to the condition $y(0) = 1$. By Picard's method of successive approximations $y^{(1)} =$ _____.

9. While evaluating the definite integral by Trapezoidal rule, the accuracy can be increased by taking _____.

10. In Lagrange Interpolation formula, the value of $L_1(x_1) =$ _____.

SECTION "B"
[10 Q. × 0.5 = 5 marks]

Fill in the blank space(s) by selecting the most appropriate answer from among the given ones. (*Do not tick the answer*).

11. If $y = 3x^7 - 6x$, then the relative error in y at $x = 1$ when error in x is 0.05, is _____.

[0.05; 0.25; 0.5; 1]

12. The number of iterations in the bisection method to find a solution of the equation $f(x) = 0$ on an interval $[0, 1]$ with a tolerance $\epsilon = 0.001$ is _____.
[10; 5; 2; 1]

13. Stirling formula for interpolation gives the most efficient result for_____.
[$-1 \leq p \leq 0$; $0 \leq p \leq 1$; $1/4 \leq p \leq 3/4$; $-1/4 \leq p \leq 1$]

14. In deriving the trapezoidal formula for the curve y =f(x) each sub- interval is replaced by its_____.
[chord; straight line; ellipse; tangent line]

15. The technique for computing the value of the function outside the given argument is called _____.
[extrapolation; interpolation; fitting; inspection]

16. Taylor's series method is used to solve _____.
[boundary value problem; real valued problem;
Initial value problem; complex valued problem]

17. The Gauss Seidel method converges approximately _____ faster than the Jacobi method.
[one times; sixty times; two times; four times]

18. Let $y(x)$ be the polynomial of nth degree with leading coefficient A_0. Then $\Delta^n y(x)$ _____ where h is the x-step size and Δ is the forward difference operator.
[0; A_0; $A_0 n!$; $A_0 n!\, h^n$]

19. In fitting a straight line, $Y = a_0 + a_1 x$, one of the two normal equations is _____.
[$ma_0 + a_1 \sum x_i = \sum y_i$; $ma_0 + \sum x_i + a_1 \sum x_i^2 = \sum x_i y_i$;
$a_0 + a_1 \sum x_i = \sum y_i$; $a_o + \sum x_i^2 = \sum x_i y_i$]

20. If $f(x)$ is continuous in (a, b) if $f(a)$ and $f(b)$ are opposite sign, then the equation $f(x) = 0$ will have at least _____ between a and b.
[one real root; two real roots; three root; four roots]

KATHMANDU UNIVERSITY
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November, 2018

NOV 15 2018

Level : B.E.
Year : II
Time : 2 hrs. 30 mins.

Course : MCSC 202
Semester : II
F.M. : 50

SECTION "C"
[6 Q. × 7 = 42 marks]

1. Derive Newton Raphson Method to find the solution of the equation, $f(x) = 0$. Also show that this method converges quadratically in the neighborhood of the root of $f(x) = 0$. Use the method to find a root of the equation $x\sin x + \cos x = 0$ correct to 4 – decimal places with the initial guess $x_0 = \pi$. [2+2+3]

2. Derive Newton divided difference interpolation formula. Using the following table find f(x) as a polynomial of x. [3+4]

x	-1	0	3	6	7
f(x)	3	-6	39	822	1611

OR

Derive Newton's forward difference interpolation formula. Use this method to determine the value of $f(0.23)$ from the following set of tabulated values:

x	0.20	0.22	0.24	0.26	0.28	0.30
$f(x)$	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

3. Derive Simpson's $\frac{1}{3}$ formula to compute the value of definite integral $I = \int_a^b y(x)dx$. Evaluate the double integral $\int_0^1 \int_0^1 e^{x+y} dx\, dy$ using Simpson's rules taking $h = k = 0.5$. [3+4]

4. Derive the formula to find the solution of initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$ by second order Runge-Kutta method. Use this method to find y at $x = 0.1, 0.2$ given that $2y' = (1 + x)y^2$ with $y(0) = 1$. [4+3]

5. Discuss LU factorization method to find the solution (x, y, z) of the system of linear equation. Use LU factorization method to find the solution correct to four decimal places. [3+4]

$$x + 2y + 3z + 4w = 10$$
$$-x + y + 2z + 3w = 5$$
$$x - y + z + 2w = 3$$
$$-x + y - z + 5w = 4$$

6. What do you mean by the least squares problem? Determine the constants a and b by the method of least square such that $y = ae^{bx}$ fit the following data. [3+4]

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

SECTION "D"

[4 Q.× 2 = 8 marks]

7. Find the general error of $u = 5xy^2/z^3$ for a given $x = 2.23$, $y = 0.45$ and $z = 9.72$ where all the values of x, y, z are correct to two decimal places.

8. (i) Round-off to four significant digits to the number: 40.0568.
   (ii) Find the precision of 4.2305.

9. Show that $\mu^2 \equiv \frac{1}{4}(\delta^2 + 4)$, where the symbols have their usual meanings.

10. A rod is rotating in a plane about one of its ends, if the following table gives the angles θ radian through with rod has turned for different values of time t seconds. Find its angular velocity at time t = 0. 2

t(in sec)	0.2	0.4	0.6	0.8
θ(in radians)	0.12	0.48	0.10	2